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Project Title: Quantum-Trained Gene Regulatory Network for Cell Fate Signaling
Client/Stakeholder: Dr. Julie Merkle, Associate Professor of Biology, University of Evansville

Project Title and Summary:

This project builds a quantum computing model that represents genes as qubits and learns regulatory relationships from real Drosophila ovary single-cell RNA-seq data. A circuit with fixed Ry encoders and 30 trainable CRY edges is trained to match the observed gene activation patterns of germline and follicle cells, and the learned edges are compared with established Gurken/EGFR biology. The goal is to recover at least 80% of known gene-gene and cell-cell interactions. Current methods can perform near random baselines and struggle with interactions between cell populations, which this project targets.

Connection to the Requirements Document:

- The goal, two-population design, Ry/CRY circuit, weighted directed graph, and accuracy evaluation all follow the requirements I wrote for Dr. Merkle.
- Two open risks from that document turned out to be real: quality control and binary activation thresholds changed my results (a labeling bug dropped about 3,200 germline cells), and the ground-truth edge list for the 80% benchmark is still not finalized.
- The "intercellular" requirement is the part I cannot fully meet with public data, because no dataset records which germline cell touches which follicle cell. Most of this period went to testing that gap.

Functional Requirements and Current Status:

- **Convert real expression data into circuit inputs —** Done. All three datasets are converted to binary activation and Ry angles.
- **Construct and train a parameterized quantum circuit —** Done. 30 directed CRY edges are trained by analytic gradient descent on KL divergence.
- **Model germline and follicle populations —** Partly done. Within each population it works on real data. Between populations it only works on synthetic pairing, because real pairing data does not exist.
- **Calculate activation probabilities and Ry angles —** Done.
- **Output a weighted, directed graph —** Done. A network diagram and edge table are saved for every run.
- **Validate with IBM Quantum hardware —** Not started. All results come from the local noiseless simulator.
- **Compare with known biology and calculate recovery accuracy —** Done, goal not yet met. The corrected FlyCellAtlas run recovered 6 of 8 known edges (75%) against the 80% target.
- **Scope gap —** the 10-gene panel is extracted for all datasets, but the circuit currently models 6 genes (grk, BicD, egl, top, kek1, spi). Orb, mago, Fas2, and Crk are not yet in the circuit.

Non-Functional Requirements and Current Status:

- **Reliability —** Partly met. Training is deterministic for a given input, but results shift with small data changes (nearly the same real cells scored 26/30 under independence and 24/30 under random synthetic contacts). No repeated or hardware runs yet.
- **Usability —** Partly met. Each run produces a network diagram and an edge table with a Correct/Incorrect column, but I have not yet reviewed them with Dr. Merkle.
- **Maintainability —** Met so far. The code is open-source Python and Qiskit, split into labeled sections with settings at the top of each script.

- **Performance —** Met. A 6-qubit simulation trains on a laptop. The 2,000-epoch loss is still falling slowly at the end, so longer training could change the values.
- **Transparency —**hetic outputs are labeled SYNTHETIC in file names and plot titles, and I report recall on known edges beside the raw score.

What I Accomplished:

- Built and ran the full pipeline (extraction, encoding, training, Qiskit execution, validation) on three public Drosophila ovary datasets.
- Found and fixed a bug where germ cell labels were missed: germline cells in FlyCellAtlas went from 1,174 to 3,267, and I added an ambient-RNA contamination filter. The score moved from 24/30 to 26/30.
- Checked all three datasets for any sample, batch, stage, or position information that could pair germline and follicle cells. None exists.
- Built a synthetic spatial test from the real FlyCellAtlas cells: 204 synthetic egg chambers with 3,264 germline-follicle contact pairs, run once with random contacts (null) and once with a planted grk-to-top coupling.

Synthetic test results (null vs planted):

	null	planted
phi grk -> top	-0.009	0.323
phi top -> grk	-0.136	0.091
strongest OTHER cross |phi|	0.146	0.148
known edges found (of 8)	5	5
validation score (of 30)	24	24

- In the planted run the grk-top coupling was strong (a top-ON follicle cell touched a grk-ON cell 91% of the time versus 56% otherwise). The trained grk to top edge reached 0.323, more than double any other cross-population edge, and was flagged. In the null run it stayed near zero.

What I Learned:

- Preprocessing matters as much as the model. A keyword bug and a contamination filter changed results more than any change to the circuit.
- The independence assumption is what makes cross-population edges unlearnable. The synthetic test shows the pipeline can detect germline-follicle coupling when pairing exists.
- The 30-edge score is a weak measure. A model that flags nothing scores 22, and my results moved by two points from small data changes, so I report found edges out of 8.
- The 75th-percentile cutoff is relative. In the planted run the new grk to top edge entered the top group and pushed spi to top just below the line, so the score stayed at 24.
- A planted link spreads to related genes. Grk also became correlated with kek1 (phi 0.105), because top and kek1 are correlated inside follicle cells, and the model cannot tell direct from indirect links.
- Coupling is symmetric, so edge direction cannot be trusted. grk to top beat top to grk (0.323 vs 0.091), but that probably reflects the circuit's structure, not biology.

Challenges:

- No public dataset pairs germline cells with the follicle cells touching them. I searched the literature and checked all three datasets and found none.
- Edge direction is ambiguous: BicD to grk was missed while grk to BicD was flagged on real data.
- Small germline populations (112 cells in GSE146040) and ambient RNA in follicle cells (about 63% above a 0.3 contamination score) limit what the data can show.
- Hardware validation and the finalized ground-truth list from Dr. Merkle are still outstanding.

Successes:

- The pipeline runs end to end on all three datasets and saves a diagram, loss curve, and edge table each time.
- Corrected real FlyCellAtlas data recovered 6 of 8 known edges, 6 of the 7 within-population ones.
- The synthetic positive control behaved as it should: no cross-population signal in the null run and a clear one in the planted run, with the same cells and the same within-population statistics in both.
- The test identified what a real dataset would need: paired germline-follicle observations.

Responsible Computing Considerations:

- Synthetic results are not biology. The planted coupling was placed there by my script, so it shows the method can detect coupling, not that Grk to EGFR coupling exists in real flies. I will not present it as a finding.
- Scoring can overstate success. I report edges recovered out of the 8 known ones beside the 30-edge score, and I note that direction is not identifiable.
- Preprocessing choices (label keywords, contamination cutoff) change results and should be documented and reviewed by the biology stakeholder.
- All datasets are public data from other research groups (NCBI GEO, FlyCellAtlas) and must be cited with their data-use terms respected.

Next Steps:

1. Explore the UCSC Cell Browser link for Drosophila ovary data (cells.ucsc.edu, dros-ovary / ovary-high-quality) to see whether it offers any real spatial information. I could not open it automatically, and its name matches the GSE146040 high-quality data I already have, so I need to check whether its layouts are tissue positions or only UMAP/t-SNE maps.
2. Make the synthetic data better: vary the coupling strength (for example 0.5 and 1.0), add distance-dependent coupling and several follicle neighbors per germline cell, and repeat across random seeds, always comparing against the null run.
3. Run the trained 6-qubit circuit on an IBM Quantum backend and compare it with the AerSimulator result, and finalize the ground-truth edge list with Dr. Merkle.